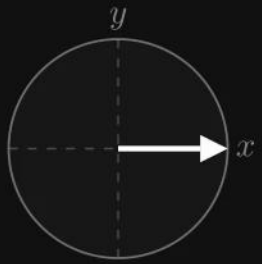
pure table against coin:
here the eigenvalue bars differ too, and that the Meter can see

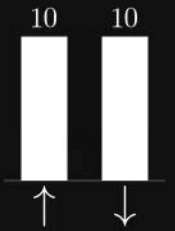


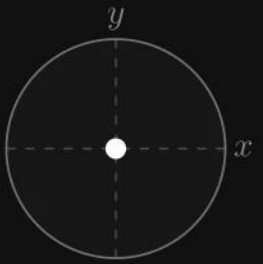
$|+x\rangle$

	↑	↓
↑	0.50	0.50
↓	0.50	0.50



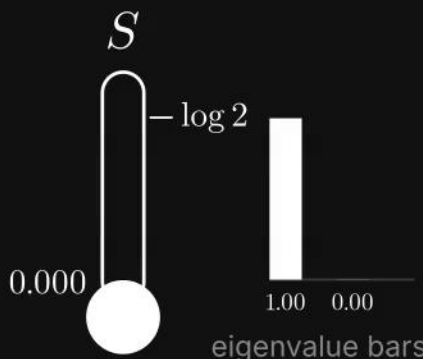
same histogram

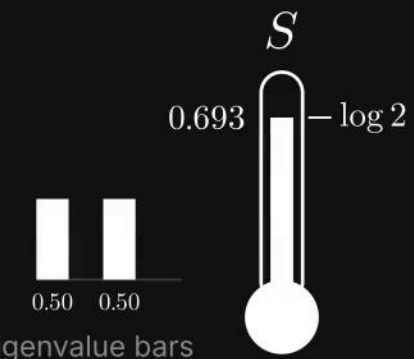




fair coin

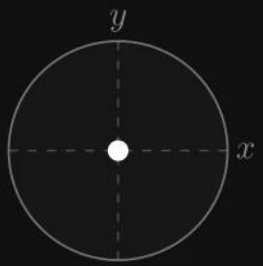
	↑	↓
↑	0.50	0.00
↓	0.00	0.50





$$\rho_{+x} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \rho_{\text{mix}} = \tfrac{1}{2}\mathbb{1}$$

a friend measures along z and does not tell

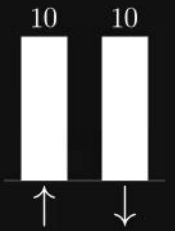


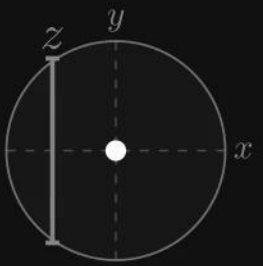
$|+x\rangle$

	↑	↓
↑	0.50	0.00
↓	0.00	0.50



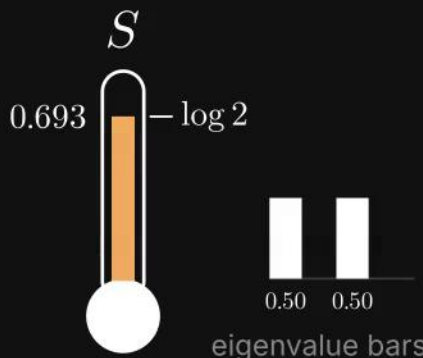
same histogram



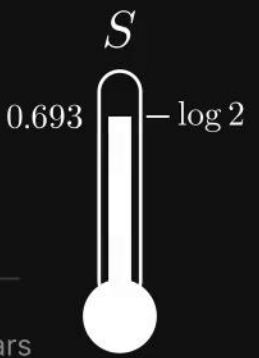


fair coin

	↑	↓
↑	0.50	0.00
↓	0.00	0.50



the secret cost $\log 2$

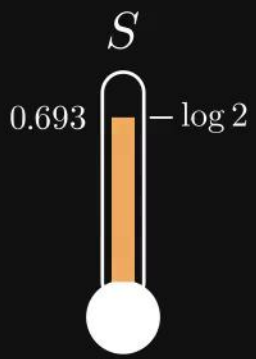


cost 0

$$\rho_{+x} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \rho_{\text{mix}} = \tfrac{1}{2}\mathbb{1}$$

2023: the asymmetry is defined (Ares, Murciano and Calabrese).
2024: microscopic origin of the quantum Mpemba effect (Rylands and co-workers); the effect seen with trapped ions (Joshi and co-workers).
2025: a review of the quantum Mpemba effects (Ares, Calabrese and Murciano).
Today: the same meter pointed at a gapped magnet, with nothing fitted.





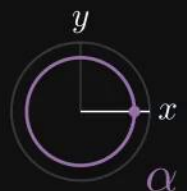
the secret cost $\log 2$

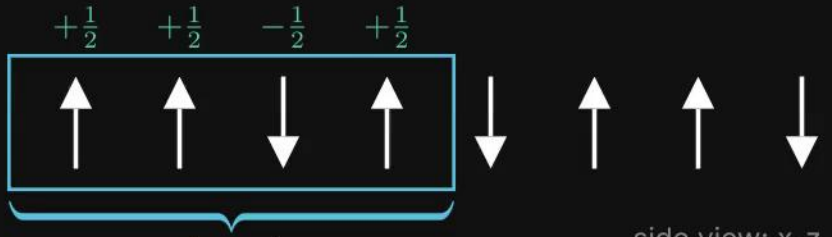
? What did the secret cost?

closed form

BLACK BOX · the 2023 definition

- The definition is taken from Ares, Murciano and Calabrese (2023) and the 2025 review; this series rebuilds its picture, not its history.
- Quantum Mpemba effect (Rylands and co-workers 2024; trapped ions, Joshi and co-workers 2024): a state that starts further from the symmetry can restore it faster after a sudden change; named here, never derived.
- The paper's own use of the meter: at equilibrium it reads how far a window sits from a symmetry that the ground state breaks.



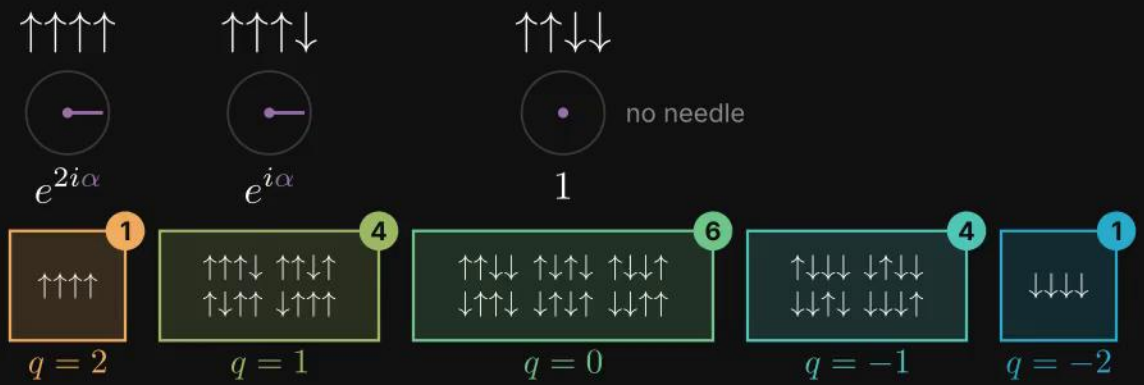


side view: x-z

ups minus downs, halved:

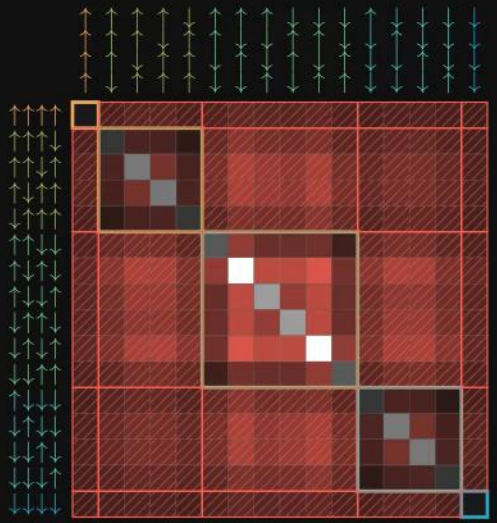
$$Q_A = 1$$

$$Q_A = \frac{1}{2}\sum_{j\in A}\sigma_j^z$$



$$e^{i\alpha Q_A}|q\rangle = e^{i\alpha q}|q\rangle$$

odd windows: half-integer q, integer differences



window $\ell = 4$ of a small magnet

(ring toy, pinned, L = 16;
$\Delta_z = 0.3,\ \gamma = 0.2$)

$q = 2$
$q = 1$
$q = 0$
$q = -1$
$q = -2$

weight of a block =
the sum of the squared
sizes of its cells

- one drawer apart: 0.347
- two apart: 0.050
- three apart: 0.003
- same drawer: 0.383

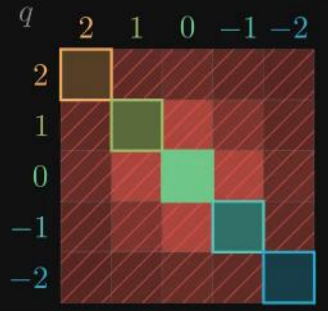
keep every block; another
state of this magnet (Part 3)
leaves the odd distances dark

$$\rho_A = \sum_{q,q'} \Pi_q \rho_A \Pi_{q'}$$

$q = q'$: the five diagonal blocks

$q \neq q'$: the red blocks

Π_q keeps what sits in drawer q

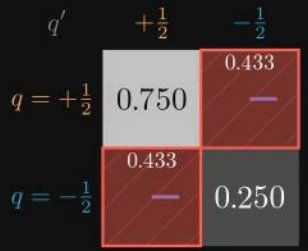


brightness: square root of the weight

ring toy, pinned, L = 16; Delta_z = 0.3, gamma = 0.2
window l = 4 (even l <= L/4)
block weights: talknum.asymmetry.block_norms

one tilted spin

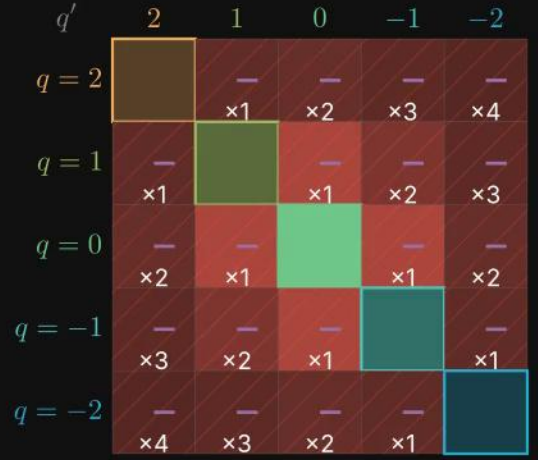




one spin: a difference of 1

colour fixed, needle turns

the 5×5 cartoon of the window $\ell = 4$

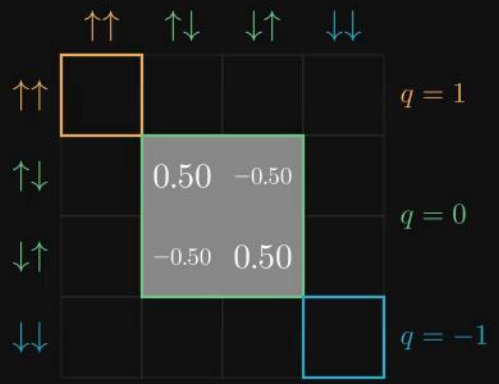


brightness: square root of the weight

$$\rho_{qq'} \mapsto e^{i\alpha(q-q')}\rho_{qq'}$$

speed = charge difference

the Bell pair, sorted by charge



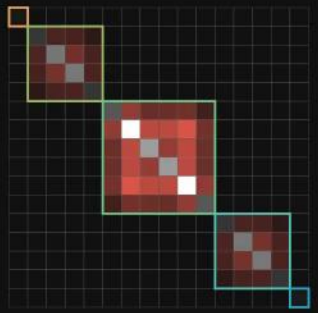
same charge, no needle to turn

PAUSE AND PONDER

A friend measures the window's charge and never tells you the result. Is the window now more certain, less certain, or the same?

Vote: More certain | Less certain | The same

a friend measures Q_A and does not tell





$q = 2$
$q = 1$
$q = 0$
$q = -1$
$q = -2$

$$\rho_{A,Q} = \sum_q \Pi_q \rho_A \Pi_q$$

$\rho_{A,Q}$: the table after the secret measurement

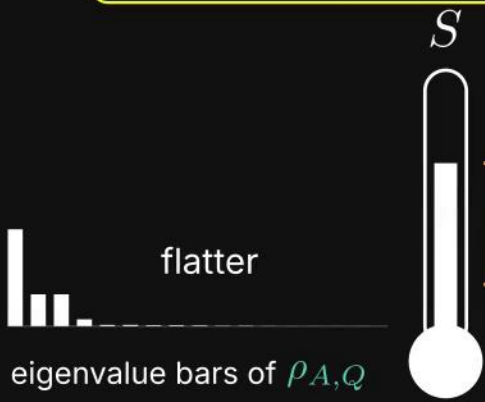
BLACK BOX · ΔS ≥ 0

- Erasing the off-block cells never lowers the Meter: ΔS ≥ 0 for every n, with equality exactly when there is nothing to erase ([ρA, QA] = 0).
- Reason, on the Ledger: the erased table is an average of rotated copies (2.8), and averaging tables never sharpens eigenvalue bars.
- The asymmetry is a difference of two Meter readings on two tables, so it can be log 2 while the Meter itself reads 0 (2.6).

? PAUSE AND PONDER

A friend measures the window's charge and never tells you the result. Is the window now more certain, less certain, or the same?

Vote: More certain | Less certain | The same

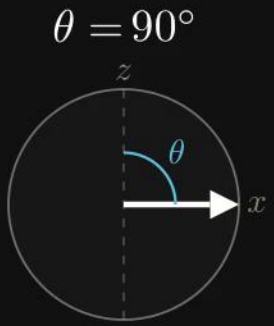


$\Delta S_A^{(2)} = 0.715$

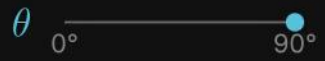
infinite chain (paper): 0.700

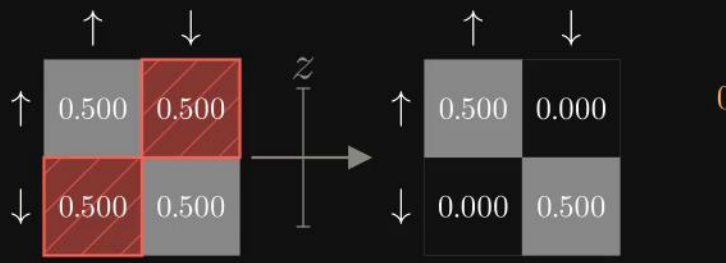
$$\Delta S_A^{(n)} = S^{(n)}(\rho_{A,Q}) - S^{(n)}(\rho_A)$$

```
ring toy, pinned, L = 16; Delta_z = 0.3, gamma = 0.2; l = 4
        Delta S^(2) = 0.715 (talknum.asymmetry.asymmetry)
paper: rdm_030_020.json n2[0], Delta S^(2)(l = 4) = 0.700
        iDMRG, infinite chain, Delta_z = 0.3, gamma = 0.2
```



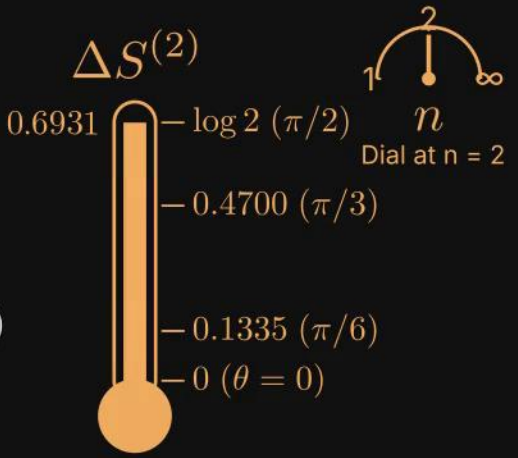
side view: x-z





secret z measurement

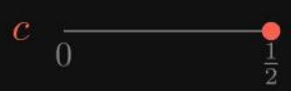
$\mathrm{diag}(\cos^2\frac{\theta}{2},\ \sin^2\frac{\theta}{2})$



$$\Delta S^{(2)} = -\log\left(\cos^4\frac{\theta}{2} + \sin^4\frac{\theta}{2}\right)$$

one spin, no partner, and the meter moves





$$\begin{pmatrix} 1/2 & c \\ c & 1/2 \end{pmatrix} \Rightarrow \log(1 + 4c^2) = 0.6931$$

a mix of the spin along x and the fair coin; c sets the proportions

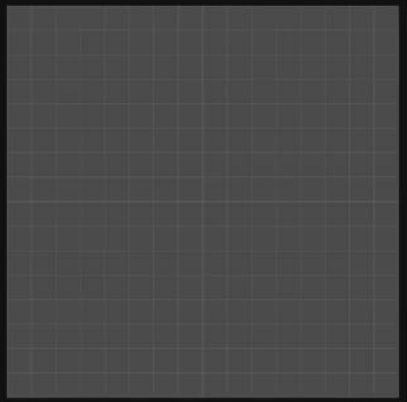
	Meter S	asymmetry
 one spin of the Bell pair	$\log 2$	0
 one spin along x	0	$\log 2$

historical name: entanglement asymmetry

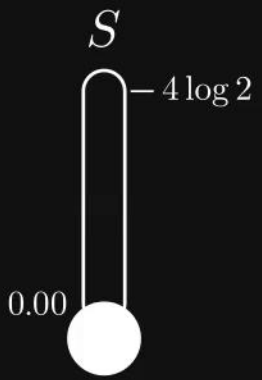
The name is historical.



the window's Grid



dense: every cell $\frac{1}{16}$



the window's Meter

PAUSE AND PONDER

Every arrow along x and the Meter at zero. As the window grows, what does the asymmetry do?

Vote: It saturates | It grows like log ℓ | It grows like ℓ



top view: x-y

$$P(q) = \binom{\ell}{\ell/2+q} 2^{-\ell}$$

$$\Delta S^{(n)} = \frac{1}{1-n} \log \sum_q P(q)^n$$

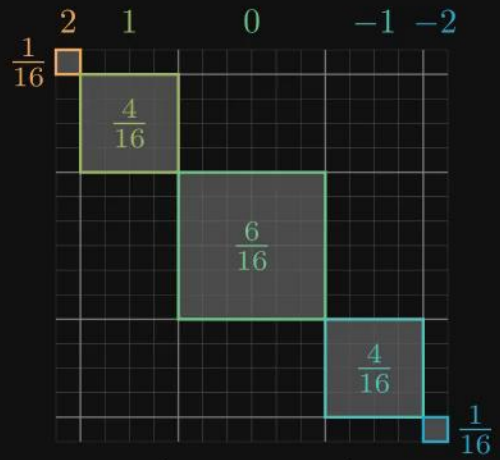
(pure)

PAUSE AND PONDER

Every arrow along x and the Meter at zero. As the window grows, what does the asymmetry do?

Vote: It saturates | It grows like log ℓ | It grows like ℓ

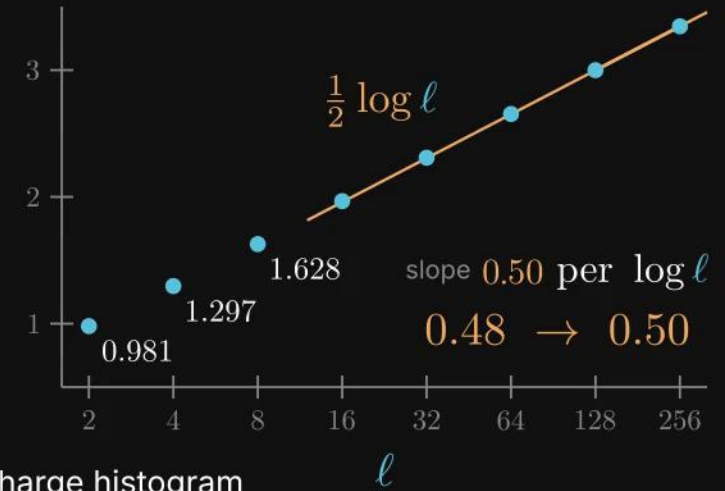
sorted by drawer q



$P(q)$



$\Delta S^{(2)}$



pure window: the eigenvalue bars of $\rho_{A,Q}$ are the charge histogram

each hand is $e^{im\alpha}$



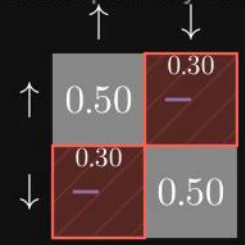
$\alpha/\pi = 1.00$

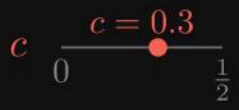
average over a full turn: 1 if m = 0, else 0

$$\Pi_q = \int_{-\pi}^{\pi} \frac{d\alpha}{2\pi}\, e^{i\alpha(Q_A - q)}$$

the drawer projector of the sorting scene (2.3)

the napkin toy of 2.6







$\alpha/\pi = 1.00$

$$\mathrm{Tr}[\rho\,\rho(\alpha)] = \tfrac{1}{2} + 2c^2\cos\alpha = 0.3200$$

circle average $= 0.5000 = \tfrac{1}{2} = \mathrm{Tr}\,\rho_{A,Q}{}^2$

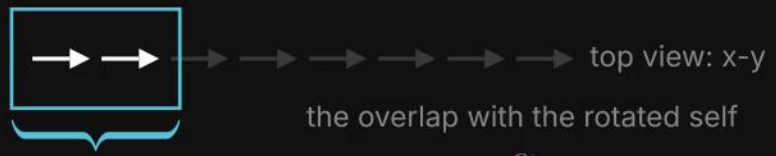
ratio, minus the log: $-\log\frac{1/2}{1/2+2c^2} = \log(1+4c^2) = 0.3075$

(the 2.6 readout)

identity checked to 10^{-6} on a random two-spin table (talknum)

$$\frac{\mathrm{Tr}\,\rho_{A,Q}{}^2}{\mathrm{Tr}\,\rho_A^2} = \int \frac{d\alpha}{2\pi}\, Z_2(\alpha)$$

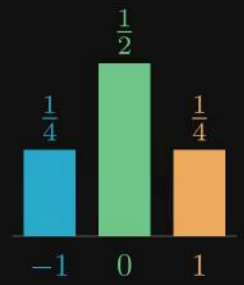
$n = 2$ is minus the log of an overlap.



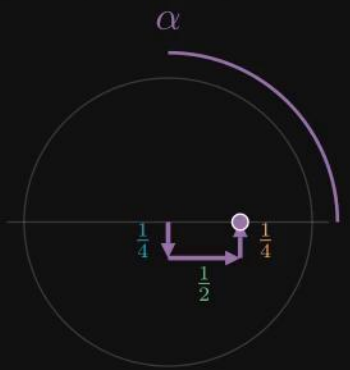
the overlap with the rotated self

$\ell = 2$

$P(q)$



charge histogram



$\alpha/\pi = +0.50$

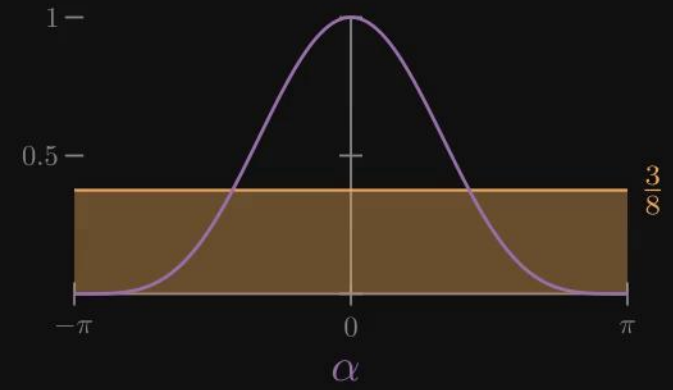
$$\sum_q P(q)\, e^{i\alpha q} = \cos^2 \tfrac{\alpha}{2}$$

$$Z_2(\alpha) = \cos^4 \tfrac{\alpha}{2}$$

$$Z_2(\alpha) = \Big|\sum_q P(q)\, e^{i\alpha q}\Big|^2$$

(pure; mixed windows in 4.2)

$Z_2(\alpha)$



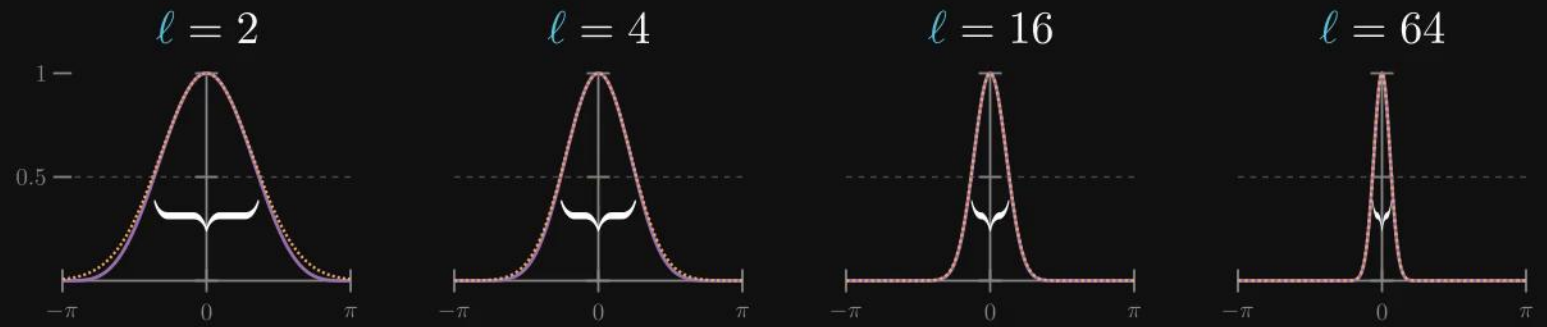
$$\int_{-\pi}^{\pi} \frac{d\alpha}{2\pi}\, Z_2(\alpha) = \tfrac{3}{8}$$

$-\log \tfrac{3}{8} = 0.981 \longrightarrow \Delta S_A^{(2)}$ at $\ell = 2$

scene 2.7, first point

$$Z_2(\alpha) = \cos^{2\ell}(\alpha/2)$$

one factor $\cos\frac{\alpha}{2}$ per spin along x, then squared



$e^{-\ell\alpha^2/4}$ Gaussian

width $\propto 1/\sqrt{\ell}$

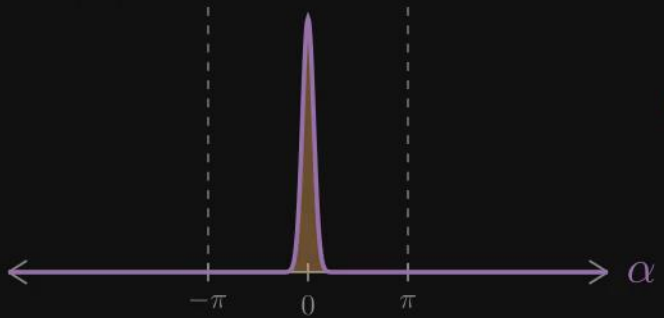
$$Z_2 \approx e^{-V\alpha^2}, \qquad V = \ell/4$$

V = the variance of the charge histogram (coins)

narrow bump in α, wide bell in q: Fourier partners

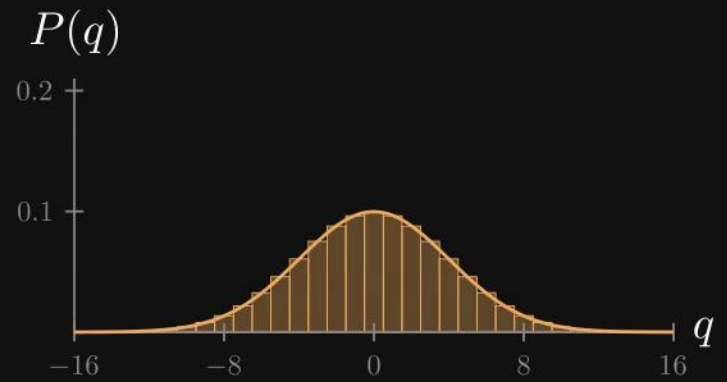
the overlap: a narrow bump in α

$Z_2(\alpha) \approx e^{-V\alpha^2}$



$\ell = 64$

the Bell, variance $V = \ell/4$

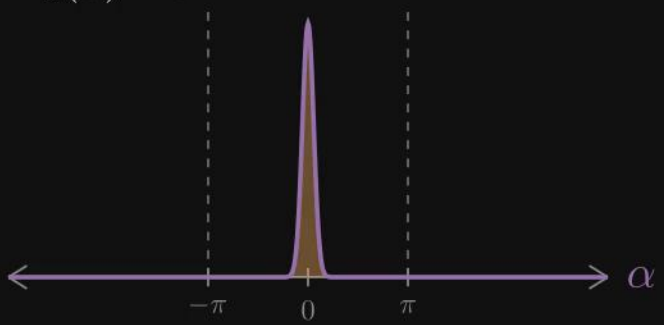


the tails we add are nothing

$$\int \frac{d\alpha}{2\pi} e^{-V\alpha^2} = \frac{1}{2\sqrt{\pi V}} \Rightarrow \Delta S_A^{(2)} = \tfrac{1}{2}\log(4\pi V) = \tfrac{1}{2}\log(\pi\ell) \text{ (coins)}$$

the overlap: a narrow bump in α

$$Z_2(\alpha) \approx e^{-V\alpha^2}$$

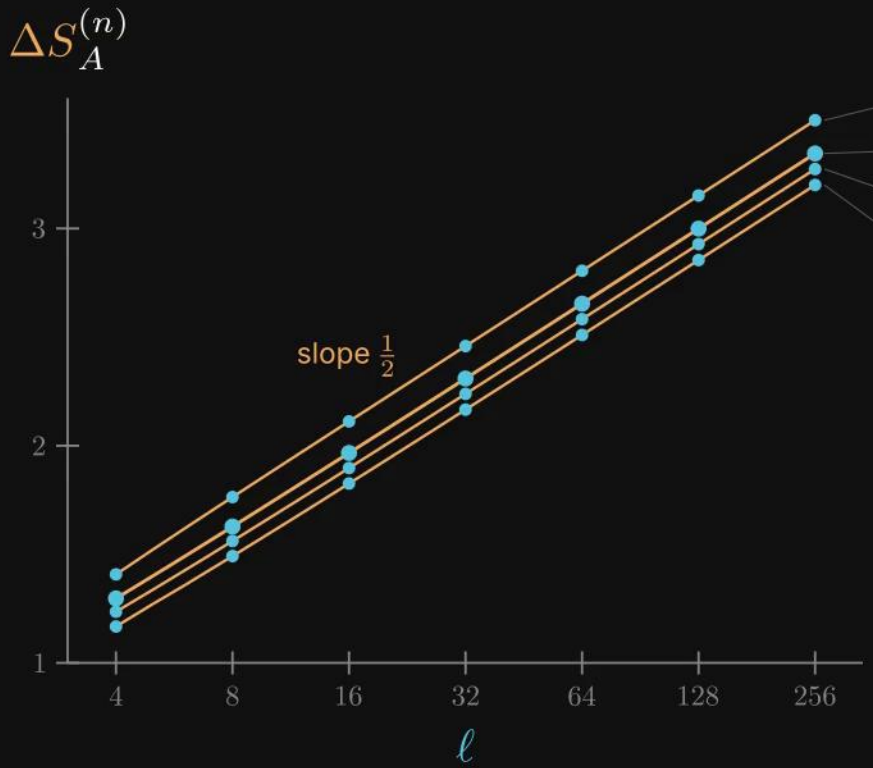


ℓ	exact	$\frac{1}{2}\log(\pi\ell)$
16	1.9665	1.9587
256	3.3454	3.3450

half the log of the variance

the tails we add are nothing

$$\int \frac{d\alpha}{2\pi} e^{-V\alpha^2} = \frac{1}{2\sqrt{\pi V}} \Rightarrow \Delta S_A^{(2)} = \tfrac{1}{2}\log(4\pi V) = \tfrac{1}{2}\log(\pi\ell) \text{ (coins)}$$



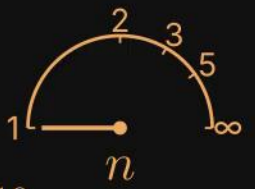
$\frac{1}{2}\log(2\pi V) +$

$n \to 1$ 0.5

$n = 2$ 0.3466

$n = 3$ 0.2747 } 0.0719

$n = 5$ 0.2012



no free parameter

$$\Delta S_A^{(n)} = \tfrac{1}{2}\log(2\pi V) + \frac{\log n}{2(n-1)}$$

closed form · talknum: coins_asymmetry, renyi_offset

$$\Delta S_A^{(n)} = \tfrac{1}{2}\log(2\pi V) + \frac{\log n}{2(n-1)}$$

BLACK BOX · replica-ring Gaussian integral

- n copies of ρ_A on a closed loop, one angle per copy, one difference per junction; if every junction's bump has the same width V, the integral is Gaussian: (1/√n)(2πV)^(−(n−1)/2).
- 1/√n is the length √n of the one mode that does not move, over the product of the other mode frequencies, ∏ 2 sin(πk/n) = n (checked in talknum).
- The paper (App. A) proves the bump has the same curvature V between any two copies, however far apart on the loop (V_d = V), for the states its numerics use; here it is taken on faith.

BLACK BOX · n → 1 continuation

- The formula is derived for whole n ≥ 2 and read at n → 1 as the limit log n / (2(n−1)) → 1/2.
- The paper confirms the n → 1 value numerically on the chain (its Fig. 2a; scene 4.4); card again in 4.3.
- The Dial of 1.8 did the same for S: the reading at n = 1 is a limit, not a separate measurement.

TAKE-HOME

$$\Delta S_A^{(n)} = S^{(n)}(\rho_{A,Q}) - S^{(n)}(\rho_A) \geq 0$$

- Sort the Grid by charge: inside a block nothing turns under a rotation about z, outside every block spins at speed q − q′; erasing the outside is what a friend's secret measurement of the charge does, and ΔS ≥ 0 is the Meter's rise.
- The asymmetry needs no entanglement: one spin along x reads log 2 with the Meter at 0; one spin of the Bell pair reads 0 with the Meter at log 2.
- The erasure is an average over all rotations (the Clock): n = 2 is minus the log of an overlap, the overlap is a Gaussian bump of width one over root ℓ, and the asymmetry is half the log of four pi V, the Bell's variance; other n differ by pure numbers.

LEDGER

- ☐ $\Delta S \geq 0$ asserted (2.5); the proof (the erased table is an average of rotated copies, a mixture of unitaries, hence flatter eigenvalue bars) is one line of majorization, not shown.
- ☐ The circle-average identity $\mathrm{Tr}\,(\rho_{\{A,Q\}})^2 = \int d\alpha/2\pi\ \mathrm{Tr}[\rho_A\, \rho_A(\alpha)]$ was checked by hand on a 2×2 table and to 10^{-6} on a random two-spin table; the general case and every $n > 2$ are App. A of the paper (2.8).
- ☐ Pure-state shortcut throughout 2.7–2.10 ($\mathrm{Tr}\,\rho_A^2 = 1$, $Z_2 = |\Sigma_q\, P(q)\, \exp(i\alpha q)|^2$); the mixed case, where V is defined by the bump's curvature and, for the paper's states (Gaussian or gapped matrix-product windows), equals the variance up to edge terms, is 4.2; for a generic mixed table the two differ (the 4.2 toy).
- ☐ The circle integral was extended to the whole line and the bump replaced by its Gaussian (2.10); the table 1.9665 / 1.9587 and 3.3454 / 3.3450 shows the size of the error.
- ☐ Replica-loop integral, “same width at every junction” and the $n \to 1$ continuation: taken on faith (2.11); the product of mode frequencies $\prod 2\sin(\pi k/n) = n$ is checked in talknum.

LEDGER

- ☐ Rotation sign: on screen the arrows turn by −α and the Grid needles by +α (q − q′), both what exp(iαQ_A) ρ_A exp(−iαQ_A) does (1.10, 2.4).
- ☐ Odd windows have half-integer charges and integer differences; every window on screen is even (2.2).
- ☐ Ring numbers are toys: ring toy, pinned, L = 16, Δ_z = 0.3, γ = 0.2, window ℓ = 4; the paper's 0.700 is an infinite-chain value and the ring never sits on the paper's curve (2.3, 2.5).
- ☐ “Entanglement asymmetry” is the historical name, said once (2.6); the quantum Mpemba effect was named, not explained (2.1).
- ☐ χ_zz was flashed, not defined; it arrives in 3.11 (2.12).

$$\Delta S_A^{(2)} = \tfrac{1}{2}\log(4\pi V), \quad V = \boxed{?}$$

χ_{zz}?

not defined yet: Part 3



Part 3: a magnet that prefers x over y

The meter reads half the log of the charge bell's variance. For coins it is ℓ/4. For interacting magnets whose Hamiltonian prefers x over y, what is the variance, who sets it, and is V still the variance when the piece is mixed?